Weibull Distribution

- Like the gamma and exponential distributions, the Weibull distribution is also applied to reliability and life-testing problems such as the **time to failure** or **life length** of a component, measured from some specified time until it fails.

Weibull Distribution

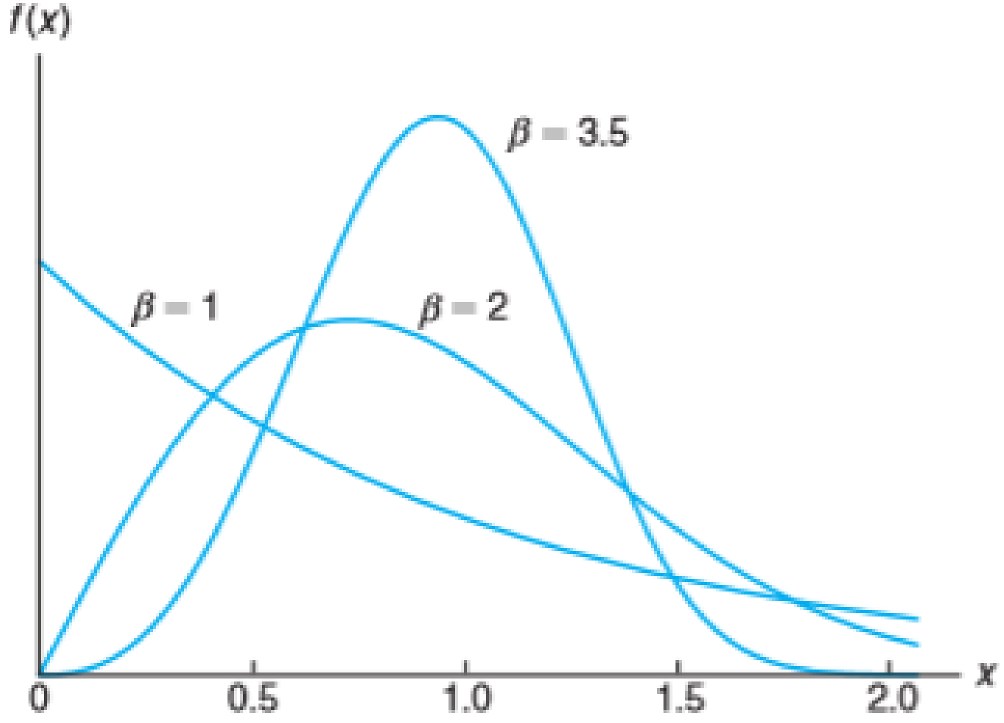
- A continuous random variable X follows weibull distribution with parameter α and β then its probability density function is given by

$$f(x,\alpha,\beta) = \begin{cases} \alpha\beta x^{\beta-1}e^{-\alpha x^{\beta}}, & x > 0 \\ 0, & otherwise \end{cases}$$

- where $\alpha > 0, \beta > 0$
- $X \sim Gamma(\alpha, \beta)$.

Observation

- When $\alpha = 1$, the graphs of weibull distribution when various values of β is given bellow.



Observation

- When $\beta = 1$, the Weibull distribution becomes exponential. $f(x) = \alpha e^{-\alpha x}$.
- The mean of X : $\alpha^{-\frac{1}{\beta}}\Gamma(1 + \frac{1}{\beta})$.
- The variance of X :

$$\alpha^{-\frac{2}{\beta}}\left\{\Gamma\left(1 + \frac{2}{\beta}\right) - \left[\Gamma\left(1 + \frac{1}{\beta}\right)\right]^2\right\}$$

Observation

- The CDF of x : F$(x) = 1 - e^{-\alpha x^{\beta}}$.
- The CDF of X : $\alpha^{-\frac{1}{\beta}}\Gamma(1 + \frac{1}{\beta})$.

- Problem 1: The length of life X, in hours, of an item in a machine shop has a Weibull distribution with α =0 .01 and β = 2. What is the probability that it fails before eight hours of usage?
- Ans : X = length of life ~Weibull(α =0 .01 , β = 2)
- The pdf of X : $f(x) = \frac{1}{100} * 2 * x^{2-1} e^{-\frac{1}{100}x^2}, x > 0$
- Q1> P(fails before eight hours of usage) = P(X<8)
- $P(X < 8) = \int_0^8 \frac{1}{100} * 2 * x^{2-1} e^{-\frac{1}{100}x^2} dx = F(8)$
- . $F(8) = 1 - e^{-\frac{1}{100}8^2}$

- Problem 2: Suppose that the service life, in years, of a hearing aid battery is a random variable having a Weibull distribution with α =1 /2 and β = 2.
  (a) How long can such a battery be expected to last?
  (b) What is the probability that such a battery will be operating after 2 years?

- Ans : X = length of life ~Weibull(α =1/2 , β = 2)
- The pdf of X : $f(x) = \frac{1}{2} * 2 * x^{2-1} e^{-\frac{1}{2}x^2}, x > 0$

a)

Lets calculate how long can a battery be expected to last.

$$
\begin{aligned}
E(X) &= \alpha^{-1/\beta}\Gamma\left(1+\frac{1}{\beta}\right) \\
&= \left(\frac{1}{2}\right)^{-1/2}\Gamma\left(1+\frac{1}{2}\right) \\
&= \sqrt{2}\frac{1}{2}\Gamma\left(\frac{1}{2}\right) \\
&= \frac{1}{\sqrt{2}}(\sqrt{\pi}) \\
&= \sqrt{\frac{\pi}{2}} \\
&\approx 1.25 \text{ years}
\end{aligned}
$$

b)

Now, lets calculate the probability that a battery will be operating after 2 years.

$$
\begin{aligned}
P(X > 2) &= 1 - P(X \leq 2) \\
&= 1 - F_X(2) \\
&\text{Using the equation (1), we get:} \\
&= 1 - \left(1 - \exp\left(-\left(\frac{1}{2}\right)(2)^2\right)\right) \\
&= 1 - 1 + \exp(-2) \\
&= e^{-2} \\
&= 0.1353
\end{aligned}
$$

- Problem 3: In the life of a certain type of a car, in years, has a Weibull distribution with $\beta = 2$.
  (a) find the value of α, given that the probability of the life of a car exceeds 5 years is $e^{-0.25}$.
  (b) Find the mean and variance.